On the risk of a cross-validated shrinkage estimator in the linear model

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Abstract

In Bayesian statistics, practitioners have recently advocated for the maximization of a cross-validated version of the expected log-predictive density as a computationally attractive alternative to the evidence for the purpose of model evaluation and hyperparameter tuning. However, no theoretical guarantees are available to support the use of this quantity. Here, I provide an explicit, non-asymptotic risk analysis of such a cross-validated estimator in the classical shrinkage problem of combining a biased and unbiased Gaussian data source (Green and Strawderman, 1991). I show that if $d \geq 5$ and $n \geq 3$, the resulting estimator uniformly dominates the baseline that doesn't use biased data at all and that it only deviates from the usual James-Stein estimator in an asymptotically negligible way. This result suggests that the leave-one-out estimator may enjoy similar frequentist properties as evidence maximization procedures.

Keywords: Bayesian statistics, expected log predictive density, leave-one-out cross-validation, shrinkage estimators, James-Stein phenomenon.

1. Introduction

Consider a Bayesian model for a parameter θ specified by a prior and observation model

$$\theta \sim p(\theta|\lambda); \quad y|\theta \sim p(y|\theta, \lambda) \tag{1}$$

and an observation model $p(y|\theta, \lambda)$. Here, θ represents the true parameter and λ a hyperparameter that we aim to select. Let $y = y_{1:n}$ be a dataset sampled from the true generating distribution $p^*(y)$. The gold-standard metric for model evaluation and selection Fong and Holmes (2020); Gelman et al. (2014); Gelfand and Dey (1994) is the evidence or marginal-likelihood

$$p(y_{1:n}; \lambda) = \int p(y_{1:n}|\theta, \lambda)p(\theta|\lambda)d\theta. \tag{2}$$

Despite of its conceptual appeal, its computation might be challenging (old computational paper), leading to a practical bottleneck: from a simulation perspective, while we can unbiasedly estimate (2) whenever samples from the prior are available, if the likelihood is peaked with respect of the prior, these estimators will suffer from large variance, scaling exponentially with the dimension. As a way to circumvent this issue the following approach has been proposed: we consider the expected log-predictive density on a new datapoint $\tilde{y} \sim p^*(\theta|\lambda)$ given our observed data and model (1):

$$\text{elpd}(y_{1:n}; \lambda) \stackrel{\text{def}}{=} \mathbb{E}_{\tilde{y}\sim p^*}\left(\log \int p(\tilde{y}|\theta)p(\theta|y_{1:n}, \lambda)d\theta\right). \tag{3}$$

The above quantity is the average of logarithmic scores (Gneiting and Raftery, 2007) of predictions over new datasets, where predictive distributions are the ones that arise from the Bayesian model specification (1). The relationship ($\mathcal{H}$ denotes the relative entropy)

$$\text{elpd}(y_{1:n}; \lambda) = \mathbb{E}_{p^*(\tilde{y})}\left(\log p^*(\tilde{y})\right) - \mathcal{H}\left(p^*(\tilde{y})|p(\tilde{y}|y_{1:n})\right)$$

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makes clear that the elpd is maximized whenever the predictive distribution is close to the the true generative mechanism p^*. In the model selection problem we must chose an optimal λ, and while the above elpd may provide an optimization rule, it corresponds to an oracle that requires access to the true p^*. We may instead consider cross validated versions of the elpd.

Definition 1
For $1 \leq p \leq n$ we define the leave-p-out expected log predictive density $\widehat{\text{elpd}}_p(y_{1:n})$, as the average (over the $\binom{n}{p}$ possible training/held-out splits) of the mean log predictive density of each held-out data per split. That is, if i indexes the number of possible splits, $\tilde{y}^{(i)}_{1:p}$ denotes a held-out set and $y^{(i)}_{1:n-p}$ the corresponding training set, then

$$\widehat{\text{elpd}}_p(y_{1:n}) = \frac{1}{\binom{n}{p}} \sum_{i=1}^{\binom{n}{p}} \frac{1}{p} \sum_{j=1}^{p} \log \int p(\tilde{y}^{(i)}_j|\theta) dp(\theta|y^{(i)}_{1:n-p}). \tag{4}$$

In particular, if $p = 1$ we obtain the leave-one-out expected log predictive density:

$$\widehat{\text{elpd}}_{loo}(y_{1:n}) = \frac{1}{n} \sum_{i=1}^{n} \log \int p(y_i|\theta) dp(\theta|y_{-i}). \tag{5}$$

The leave-one-out version of elpd, $\widehat{\text{elpd}}_{loo}$ has been advocated as a practical, computationally efficient estimate of the elpd (Vehtari et al., 2017, 2018), in turn, providing us with an alternative to marginal likelihood maximization. The main advantage of the elpd is that it can be efficiently approximated. Because of this, it has enjoyed much attention and is currently implemented in the main computational pipelines. Unfortunately, regardless of its wide use $\widehat{\text{elpd}}_{loo}$, little is known about its theoretical performance. If anything, classical results on leave-one-out cross validation suggest that $\widehat{elpd}_{loo}$ might suffer from inconsistent behavior.

Recently, Fong and Holmes (2020) established the remarkable relationship that the log marginal likelihood decomposes as the sum of all possible cross-validated versions of the elpd, i.e. that

$$\log p(y_{1:n}) = \sum_{p=1}^{n} \widehat{\text{elpd}}_p \left(y_{1:n}\right).$$

The above decomposition suggest that the marginal likelihood is more informative than any of the cross-validated elpd's, but in our model selection task it is unclear how each $\widehat{elpd}_p$ contributes an informative signal on λ. In this work we focus on a classical Shrinkage problem for which a risk analysis of $\widehat{\text{elpd}}_p$ is available, and we are able to establish risk bounds that are only slightly worse than the marginal likelihood baseline.

1.1. Combining biased and unbiased data

We consider the classical problem of incorporating a biased and unbiased data-sources. In this section we elaborate the specialized application in the Gaussian case, and defer the more general formulation in terms of modular inference to the discussion. We state the main result in the more general framework of the classical risk analysis of shrinkage estimators, where the data combination problem is only an instance. Let $y_i \sim \mathcal{N}(\theta, I_d\sigma_y^2)$ be an i.i.d sample in $\mathbb{R}^d$ and $z \in \mathbb{R}^d$ be a fixed vector. We consider estimators of the form

$$\hat{\theta} = \lambda\bar{y} + (1-\lambda)z = z + \lambda(\bar{y} - z). \tag{6}$$

these estimators can be understood as posterior means on a Bayesian model (1) that is specified by an independent Gaussian prior on θ centered at z and variance hyperparameter σ^2, with the assignment

$$\lambda = \frac{n\sigma^2}{\sigma_y^2 + n\sigma^2}, \quad \text{or} \quad \sigma^2 = \frac{\sigma_y^2}{n}\left(\frac{\lambda}{1-\lambda}\right). \tag{7}$$

From equation (7), we observe that the problem of finding a data-dependent λ (i.e. $\hat{\lambda}$) is equivalent to the one finding an estimate $\hat{\sigma}^2$ for σ^2. For example, maximizing the marginal likelihood of the underlying Bayesian model leads to the usual Empirical Bayes (EB) estimates

$$\hat{\sigma}^2_{EB} = \frac{\|\bar{y} - z\|^2}{d} - \frac{\sigma_y^2}{n} \quad \hat{\lambda}_{EB} = 1 - \frac{\sigma_y^2 d}{n\|\bar{y} - z\|^2}. \tag{8}$$

Although the above estimators are always well-defined, we might also want to properly represent that σ^2 is a positive quantity. Explicitly enforcing such constraints lead to the following positive part estimators

$$\hat{\sigma}^2_{EB,+} = \left(\frac{\|\bar{y} - z\|^2}{d} - \frac{\sigma_y^2}{n}\right)_+ \quad \hat{\lambda}_{EB,+} = \left(1 - \frac{\sigma_y^2 d}{n\|\bar{y} - z\|^2}\right)_+. \tag{9}$$

These estimators were originally introduced in Green and Strawderman (1991) from a shrinkage perspective. There, bypassing the need to elaborate on a Bayesian model, it was shown that choices of lambda similar to λ_{EB} and λ^+_{EB}.

2. A Risk bound for the cross-validated leave one out estimator in the Gaussian case

Here, instead of maximizing the evidence, we consider the optimization of cross-validated versions of the elpd. We do so in the following way: for a given p we will maximize the cross-validated elpd (4) with respect to λ (equivalently, σ). We will call this estimator $\hat{\lambda}_p$. This estimator gives rise to an optimal weighting for (6). We call the resulting estimator $\hat{\lambda}_{loo}$. We present an in-depth risk analysis for the estimator $\hat{\theta}_{loo}$ and give conditions for it to improve uniformly (on b) over the 'cut' estimator, the empirical mean.

The first observation is that under this model specification, simple, closed-form expressions are available for the *leave-p-out* empirical version of the elpd, as stated below

Proposition 1
Consider the the negative leave-p-out objective (4) *for the model* (**??**)*. This loss can be written as*

$$\begin{aligned} -2\widehat{elpd}_p(y;z) \;=\; & \frac{n-p}{\sigma_y^2(n-p+\lambda_p)}\lambda_p^2\left[\|\bar{y}-\bar{z}\|^2 + \frac{pS_y}{n(n-p)}\right] + \\ & -2\frac{n-p}{\sigma_y^2(n-p+\lambda_p)}\lambda_p\left[\|\bar{y}-\bar{z}\|^2 - \frac{S_y}{n}\right] + \\ & \frac{n-p}{\sigma_y^2(n-p+\lambda_p)}\left[\|\bar{y}-\bar{z}\|^2 + \frac{(n-1)S_y}{n}\right] + d\log\left(\sigma_y^2\left(1+\frac{\lambda_p}{n-p}\right)\right), \end{aligned} \tag{10}$$

where λ_p is linked to σ^2, the prior variance of θ through

$$\sigma^2 = \frac{\sigma_y^2}{n-p}\frac{\lambda_p}{1-\lambda_p}, \tag{11}$$

i.e., the same as (7) *but with $n-p$ instead of n.*

The above expression is simple enough so that we we can explicitly derive an expression for $\hat{\lambda}_p$, as shown in the following proposition.

Proposition 2
Define

$$\Delta = \|\bar{y} - z\|^2, \quad \tilde{\Delta} = \Delta + \frac{p}{n(n-p)} S_y,$$

The minimizer of the leave-p-out objective $\widehat{\text{elpd}}_p$ *writes as* $\hat{\lambda}_p^+$*, the positive part of* $\hat{\lambda}_p$ *defined below:*

$$\hat{\lambda}_p = \frac{\sqrt{4\gamma_{n,p}\Delta^2 + 4S_y^2\kappa_{n,p}(\kappa_{n,p}-1) + 4S_y\gamma_{n,p}\Delta(2\kappa_{n,p}-1) + d^2\sigma_y^4} - (2(n-p)^2\tilde{\Delta} + d\sigma_y^2)}{2(n-p)\tilde{\Delta}} \tag{12}$$

where

$$\kappa_{n,p} = \frac{p(n-p+1)}{n}, \gamma_{n,p} = (n-p)(n-p+1)$$

By Proposition 2, the negative *leave-p-out* objective (10) admits a unique minimizer $\hat{\lambda}_p^+$, the solution to the quadratic equation (12). By equation (11), this $\hat{\lambda}_p^+$ is associated with a unique optimal $\hat{\sigma}$. In turn, by equation (7) this $\hat{\sigma}$ is in correspondence to the to the final plug-in $\hat{\lambda}_{loo}^+$ to be used in (6). By simple manipulations based on (11), the relationship between $\hat{\lambda}_{loo}^+$ and $\hat{\lambda}_p^+$ is given by

$$\hat{\lambda}_{loo}^+ = \frac{n\hat{\lambda}_p^+}{(n-p) + p\hat{\lambda}_p^+} = 1 - (n-p)\frac{1-\hat{\lambda}_p^+}{n - p(1-\hat{\lambda}_p^+)}. \tag{13}$$

Notably, this estimator must be positive by construction, as opposed to the empirical Bayes case where the positivity constraint is imposed to obtain a positive $\hat{\sigma}$. To estimate the risk of this estimator we will rely on somehow classical expressions for the risk.

Proposition 3
Let $Y \sim \mathcal{N}(\theta, \Sigma)$ *where* $\Sigma = I_d\sigma^2$ *and a vector* $z \in \mathbb{R}^d$*. Consider an estimator of the form*

$$\hat{\theta} = z + \left(1 - \frac{c(\|Y-z\|^2)}{\|Y-z\|^2}\right)(Y-z), \tag{14}$$

where $c(\cdot)$ *is a differentiable function. Its risk satisfies*

$$R(\hat{\theta}, \theta) = d\sigma^2 + E\left(\frac{c^2\left(\|Y-z\|^2\right)}{\|Y-z\|^2}\right) + \sigma^2 E\left(2(2-d)\frac{c\left(\|Y-z\|^2\right)}{||Y-z||^2}\right) - 4\sigma^2 E\left(c'\left(\|Y-z\|^2\right)\right) \tag{15}$$

Based on the above expression we can state the our main result, giving an explicit risk bound for the resulting estimator θ_{loo}.

Theorem 1
If $p = 1$*, the loo-CV estimator satisfies the risk bound* $\hat{\lambda}_{loo}^+$

$$\begin{aligned} R(\hat{\theta}_{loo}, \theta) &= R(\hat{\theta}_{EB}, \theta) + \frac{\sigma_y^4}{n^2} E\left(\left(\frac{2d}{n-1} + \frac{8\sqrt{2d}}{27\sqrt{n-1}}\right)\frac{1}{\|\bar{Y}-z\|^2}\right) \\ &= \frac{d\sigma_y^2}{n} + \frac{\sigma_y^4}{n^2} d(4-d) E\left(\frac{1}{\|\bar{Y}-z\|^2}\right) + \frac{\sigma_y^4}{n^2} E\left(\left(\frac{2d}{n-1} + \frac{8\sqrt{2d}}{27\sqrt{n-1}}\right)\frac{1}{\|\bar{Y}-z\|^2}\right). \end{aligned}$$

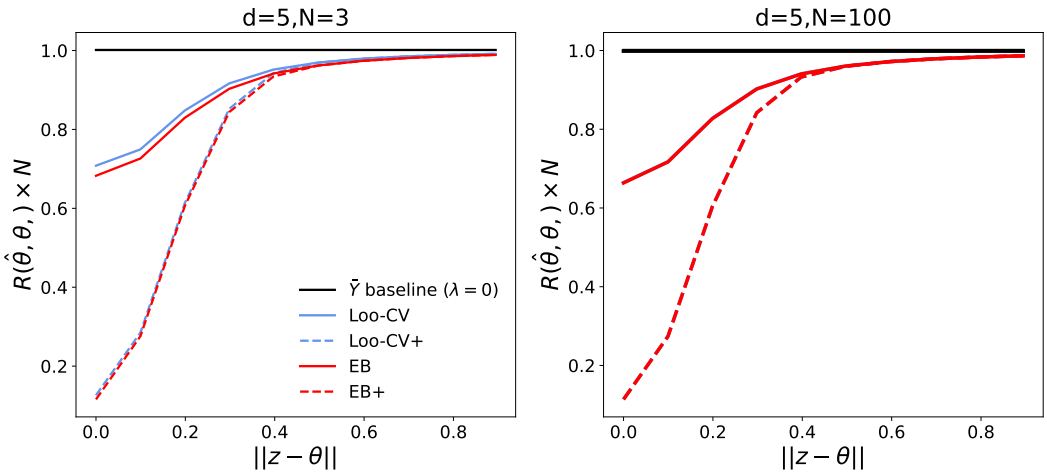

Figure 1: Risk of empirical bayes (EB) and leave-one-out cv estimator and their positive parts, for $d = 5$ and two values of N.

In particular, it has lower risk than the sample average $\hat{\theta}_0 = \bar{Y}$, if $d \geq 5$ and $n \geq 3$.

3. Simulations

In Figure 1 we compare the performance of the two estimators that we described for $d = 5$. As predicted by Theorem 1 they both dominate the sample average baseline (black line) regardless of the value of z. As also predicted by 1, both estimators become indistinguishable as N grows. As expected, gains from including biased data are greatest if the bias is small, i.e. if $||z - \theta||$ is small. As this quantity increases the risk becomes similar to the one of $\bar{Y}$ meaning that adding biased data doesn't help (but it does not harm either, in terms of the risk).

Proof. We will bound the risk of the the non truncated estimator $\hat{\lambda}_{loo}$. Based on classical results Baranchik (1964); Fourdrinier et al. (2018) we conclude that the risk can only decrease by replacing $\hat{\lambda}_{loo}$ by its truncated version $\hat{\lambda}^{+}_{loo}$. Recall that, by equation (6), our estimator $\hat{\theta}_{loo,s}$ is of the form

$$\hat{\theta}_{loo,s} = z + \hat{\lambda}_{loo,s}(\bar{Y} - z),$$

where $\hat{\lambda}_{loo,s}$ is given by equations (12) and (13). This $\hat{\lambda}_{loo,s}$ depends on data through the independent quantities $||\bar{Y} - z||^2$ and S_y. Then, conditional on S_y, $\hat{\theta}_{loo,s}$ can be expressed as in (14) for some c_{loo}, so that (15) in Proposition (3) holds conditional on S_y. Using the law of iterated expectation, we obtain that (15) holds unconditionally for a certain c_{loo} that depends on S_y and $\Delta = ||\bar{Y} - z||^2$. To compute the risk we apply Proposition (3) using the fact that $\bar{Y}$ has a multivariate Gaussian distribution with mean θ and (diagonal) variance σ_y^2/n. We will pivot on the empirical Bayes estimator $\hat{\lambda}_{EB}$ in equation (8), which corresponds to constant cost function $c_{EB}(x) = d\sigma_y^2/n$, and has a simpler risk expression. Indeed, by (15)

$$\begin{aligned} R(\hat{\theta}_{EB}, \theta) &= \frac{d\sigma_y^2}{n} + \left(\frac{d^2\sigma_y^4}{n^2} + \frac{\sigma_y^2}{n} \left(2(2-d)\frac{d\sigma_y^2}{n} \right) \right) E\left(\frac{1}{||\bar{Y} - z||^2} \right) \\ &= \frac{d\sigma_y^2}{n} + \frac{\sigma_y^4}{n^2} d(4-d) E\left(\frac{1}{||\bar{Y} - z||^2} \right). \end{aligned}$$

Now, call $c_\delta(x) = c_{loo}(x) - c_{EB}(x)$. Then,

$$\begin{aligned} R(\hat{\theta}_{loo}, \theta) &= R(\hat{\theta}_{EB}, \theta) + E\left(\frac{c_\delta^2\left(\|\bar{Y} - z\|^2\right)}{\|\bar{Y} - z\|^2}\right) + 2E\left(\frac{(c_{EB}c_\delta)\left(\|\bar{Y} - z\|^2\right)}{\|\bar{Y} - z\|^2}\right) \\ &\quad + \frac{\sigma_y^2}{n} E\left(\frac{(2(2-d)c_\delta\left(\|\bar{Y} - z\|^2\right)}{\|\bar{Y} - z\|^2}\right) - 4\frac{\sigma_y^2}{n} E\left(c_\delta'\left(\|\bar{Y} - z\|^2\right)\right) \\ &= \frac{d\sigma^2}{n} + \frac{\sigma_y^4}{n^2} d(4-d) E\left(\frac{1}{\Delta}\right) + E\left(\frac{c_\delta^2(\Delta) + 4\frac{\sigma_y^2}{n}\left(c_\delta(\Delta) - \Delta c_\delta'(\Delta)\right)}{\Delta}\right). \end{aligned} \tag{16}$$

We will work out more explicit expressions for c_δ and its derivative so that we can bound (16). We show that

$$c_\delta(x) = \frac{1}{2n}\left(R - d\sigma_y^2 - 2\gamma_{n,p}x\right), \tag{17}$$

and

$$c_\delta'(x) = \frac{2\gamma_{n,p}}{2n}\left(\frac{2\gamma_{n,p}x + Sy}{R} - 1\right), \tag{18}$$

so that

$$c_\delta(x) - xc_\delta'(x) = \frac{1}{2n}\left(\frac{2x\gamma_{n,p}S_y + d^2\sigma^4}{R} - d\sigma^2\right), \tag{19}$$

where

$$R = R(x) = \sqrt{4\gamma_{n,p}^2 x^2 + 4\gamma_{n,p}S_y x + d^2\sigma_y^4}$$

is the square root term in (12) seen as a function of $x = \Delta = \|\bar{Y} - z\|^2$. Indeed, first, by equations (12) and (13) we have

$$1 - \hat{\lambda}_p = \frac{2\gamma_{n,p}\tilde{\Delta} + d\sigma_y^2 - R}{2(n-p)\tilde{\Delta}}.$$

Therefore,

$$\begin{aligned} \hat{\lambda}_{loo} &= 1 - (n-p)\frac{1 - \hat{\lambda}_p}{n - p(1 - \hat{\lambda}_p)} \\ &= 1 - (n-p)\frac{2\gamma_{n,p}\tilde{\Delta} + d\sigma_y^2 - R}{2n(n-p)\tilde{\Delta} - p(2\gamma_{n,p}\tilde{\Delta} + d\sigma_y^2 - R)} \\ &= 1 - (n-p)\frac{2\gamma_{n,p}\tilde{\Delta} + d\sigma_y^2 - R}{2n(n-p)(1-\kappa_{n,p})\tilde{\Delta} + p(R - d\sigma_y^2)} \end{aligned}$$

Now, since $p = 1$ we have that $\gamma_{n,p} = n(n-1), \kappa_{n,p} = 1$ and $\tilde{\Delta} = \Delta + S_y/\gamma_{n,p}$, so

$$\begin{aligned}
\hat{\lambda}_{loo} &= 1 - (n-1)\left(\frac{2\gamma_{n,p}\tilde{\Delta}}{R - d\sigma_y^2} - 1\right) \\
&= 1 - (n-1)\left(\frac{2\gamma_{n,p}\tilde{\Delta}(R + d\sigma_y^2)}{(R^2 - d^2\sigma_y^4)} - 1\right) \\
&= 1 - (n-1)\left(\frac{2\gamma_{n,p}\tilde{\Delta}(R + d\sigma_y^2)}{4\Delta\gamma_{n,p}(\Delta\gamma_{n,p} + S_y)} - 1\right) \\
&= 1 - (n-1)\left(\frac{R + d\sigma_y^2}{2n(n-1)\Delta} - 1\right) \\
&= 1 - \frac{1}{2n\Delta}\left(R + d\sigma_y^2 - 2n(n-1)\Delta\right).
\end{aligned}$$

Then, $c_{loo}(x) = (R(x) + d\sigma_y^2 - 2\gamma_{n,p}x)/(2n)$. We obtain $c_\delta(x)$ in (17) by subtracting $c_{EB}(x)$. The derivative $c'_\delta(x)$ in (18) follows by differentiation with respect to x, using that $R'(x) = 2\gamma_{n,p}(2\gamma_{n,p}x + Sy)/R(x)$.

We also can evaluate the difference $c_\delta(x) - xc'_\delta(x)$ to obtain (19):

$$\begin{aligned}
c_\delta(x) - xc'_\delta(x) &= \frac{1}{2n}\left(R - d\sigma_y^2 - 2\gamma_{n,p}x - 2x\gamma_{n,p}\left(\frac{2\gamma_{n,p}x + S_y}{R} - 1\right)\right) \\
&= \frac{1}{2n}\left(\frac{R^2 - 4\gamma_{n,p}^2x^2 - 2x\gamma_{n,p}S_y}{R} - d\sigma_y^2\right) \\
&= \frac{1}{2n}\left(\frac{2x\gamma_{n,p}S_y + d^2\sigma_y^4}{R} - d\sigma_y^2\right).
\end{aligned}$$

We have established (18) and (19). From these, we will bound (16). We will do so by conditioning on Δ and using the independence of S_y from Δ. are independent. First, let's find a more convenient expression for $c_\delta(x)$. We use now the fact that

$$R(x) = \sqrt{(2\gamma_{n,p}x + d\sigma_y^2)^2 + 4x\gamma_{n,p}(S_y - d\sigma_y^2)}$$

to further evaluate (calling $g(x) = 2\gamma_{n,p}x + d\sigma_y^2$ and $h(x) = 4\gamma_{n,p}x(S_y - d\sigma_y^2)$)

$$\begin{aligned}
c_\delta(x) &= \frac{1}{2n}\left(\sqrt{g(x)^2 + h(x)} - g(x)\right) \\
&= \frac{1}{2n}\left(\frac{h(x)}{\sqrt{g(x)^2 + h(x)} + g(x)}\right) \\
&= \frac{1}{2n}\left(\frac{4\gamma_{n,p}x(S_y - d\sigma_y^2)}{R + 2\gamma_{n,p}x + d\sigma_y^2}\right)
\end{aligned}$$

With this, we can estimate its second moment. Indeed, using that $R + d\sigma_y^2 > 0$

$$\begin{aligned}
E\left(\frac{c_\delta^2(\Delta)}{\Delta}\right) &= E\left(E\left(\frac{c_\delta^2(\Delta)}{\Delta}\middle|\Delta\right)\right) \\
&= \frac{1}{4n^2}E\left(\frac{1}{\Delta}E\left(\frac{16\Delta^2\gamma_{n,p}^2(S_y - d\sigma^2)^2}{(R + 2\gamma_{n,p}\Delta + d\sigma_y^2)^2}\middle|\Delta\right)\right) \\
&\leq \frac{1}{n^2}E\left(\frac{1}{\Delta}E\left((S_y - d\sigma^2)^2\middle|\Delta\right)\right) \\
&\leq \frac{2\sigma_y^4 d}{n^2(n-1)}E\left(\frac{1}{\Delta}\right). \qquad (20)
\end{aligned}$$

In the last line, we used that $Var(\chi^2(f)) = 2f$. Likewise, we have the following estimate

$$\begin{aligned}
E\left(c_\delta(\Delta) - \Delta c_\delta'(\Delta)|\Delta\right) &= \frac{1}{2n}\left(E\left(\frac{2\Delta\gamma_{n,p}S_y}{R}\middle|\Delta\right) - d\sigma_y^2 + E\left(\frac{d^2\sigma^4}{R}\middle|\Delta\right)\right) \\
&\leq \frac{1}{2n}\left(E\left(\frac{2\gamma_{n,p}\Delta d\sigma_y^2}{2\gamma_{n,p}\Delta + d\sigma_y^2}\middle|\Delta\right) - d\sigma_y^2 + E\left(\frac{d^2\sigma^4}{R}\middle|\Delta\right)\right) \\
&\leq \frac{d^2\sigma_y^4}{2n}E\left(\frac{1}{R} - \frac{1}{2\gamma_{n,p}\Delta + d\sigma_y^2}\middle|\Delta\right).
\end{aligned}$$

In the above display, we used the conditional version of Jensen's inequality, the fact that $E(S_y|\Delta) = d\sigma_y^2$ and the concavity of the function

$$f(S_y) = \frac{S_y}{\sqrt{(2\gamma_{n,p}\Delta + \sigma_y^2)^2 + 4\Delta\gamma_{n,p}(S_y - d\sigma_y^2)}}.$$

To obtain a more useful bound, note that by multiplying numerator and denominator above by $R + (2\gamma_{n,p}\Delta + d\sigma_y^2) \leq 2(2\gamma_{n,p}\Delta + d\sigma_y^2)$ to obtain

$$\begin{aligned}
E\left(c_\delta(\Delta) - \Delta c_\delta'(\Delta)|\Delta\right) &\leq \frac{d^2\sigma_y^4}{2n}E\left(\frac{4\Delta\gamma_{n,p}(d\sigma_y^2 - S_y)}{R(2\gamma_{n,p}\Delta + d\sigma_y^2)(R + (2\gamma_{n,p}\Delta + d\sigma_y^2)}\middle|\Delta\right) \\
&\leq \frac{d^2\sigma_y^4}{2n}\frac{2\Delta\gamma_{n,p}}{(2\gamma_{n,p}\Delta + d\sigma_y^2)^3}E\left(|d\sigma_y^2 - S_y|\middle|\Delta\right) \\
&\leq \frac{d^2\sigma_y^4}{2n}\frac{2\gamma_{n,p}}{8\gamma_{n,p}^3}\frac{\Delta}{\left(\Delta + d\sigma_y^2/(2\gamma_{n,p})\right)^3}\frac{\sigma_y^2\sqrt{2d}}{\sqrt{n-1}} \\
&\leq \frac{d^2\sigma_y^4}{8n\gamma_{n,p}^2}\frac{16}{27}\left(\frac{\gamma_{n,p}^2}{d^2\sigma_y^4}\right)\frac{\sigma_y^2\sqrt{2d}}{\sqrt{n-1}} \\
&\leq \frac{2\sqrt{2d}\sigma_y^2}{27n\sqrt{n-1}}. \qquad (21)
\end{aligned}$$

Above, we have used Cauchy–Schwarz inequality and the fact that the function $f(x) = x/(x+A)^3$ achieves its maximum at $x = A/2$ if $A > 0$. We conclude by replacing (21) and (20) in (16)

□

A. Proofs

A.1. Proof of Proposition 1

Proof. For each p consider all possible training/test sets of size $n-p$ and p as $(y^{(i)}_{1:n-p}, \tilde{y}^i)$ indexed by i. For an an arbitrary training set $y^{(i)}_{1:n-p}$ denote $\bar{y}_i$ its average. The sufficient statistic $\bar{y}_i$ has a normal distribution with mean $\bar{y}_i$ and variance $\sigma^2/(n-p)$. By simple Gaussian conjugacy, the posterior distribution $p(\theta|y^{(i)}_{1:n-p})$ is normal with posterior mean μ_{post} and variance σ^2_{post} where

$$\mu_{post} = \lambda_p \bar{y}_i + (1-\lambda_p)z, \quad \sigma^2_{post} = \frac{1}{\dfrac{n-p}{\sigma_y^2} + \dfrac{1}{\sigma^2}} = \lambda_p \frac{\sigma_y^2}{(n-p)}. \tag{22}$$

Therefore, the predictive distribution is Gaussian with mean $\mu_{pred} = \mu_{post}$ and variance

$$\sigma^2_{pred} = \sigma_y^2 + \sigma^2_{post} = \sigma_y^2 \left(1 + \frac{\lambda_p}{n-p}\right).$$

And then, the log predictive density evaluated at each held-out out observation $\tilde{y}_j$ expresses as

$$\begin{aligned}
\log p(\tilde{y}^{(i)}_j | y^{(i)}_{1:n-p}) &= -\frac{1}{2} d \log\left(\sigma_y^2\left(1+\frac{\lambda_p}{n-p}\right)\right) - \frac{1}{2}\frac{n-p}{\sigma_y^2(n-p+\lambda_p)}\|\tilde{y}^{(i)}_j - (\lambda_p\bar{y}_i + (1-\lambda_p)z)\|^2 \\
&= -\frac{1}{2} d \log\left(\sigma_y^2\left(1+\frac{\lambda_p}{n-p}\right)\right) \\
&\quad -\frac{1}{2}\frac{n-p}{\sigma_y^2(n-p+\lambda_p)}\left(\lambda_p^2\|\bar{y}_i - z\|^2 - 2\lambda_p\left(\bar{y}_i - z\right)\cdot\left(\tilde{y}^{(i)}_j - z\right) + \|\tilde{y}^{(i)}_j - z\|^2\right).
\end{aligned}$$

To obtain the *leave-p-out* loss (10), we must average the above expression over all possible data spits and over the p components of $\tilde{y}^i$. Note that (except for data-independent terms) this expression is a quadratic function where each of the quadratic, linear, and constant terms are data dependent. Therefore, it suffices to take the average over each of the terms accompanying these coefficients.
Quadratic coefficient we have

$$\begin{aligned}
\sum_{i=1}^{\binom{n}{p}}\sum_{j=1}^{p}\|\bar{y}_i - z\|^2 &= p\left(\sum_{i=1}^{\binom{n}{p}}\|\bar{y}_i\|^2 - 2\sum_{i=1}^{\binom{n}{p}}\bar{y}_i\cdot z + \binom{n}{p}\|z\|^2\right) \\
&= p\left(\sum_{i=1}^{\binom{n}{p}}\|\bar{y}_i\|^2 - 2\binom{n}{p}\bar{y}\cdot z + \binom{n}{p}\|z\|^2\right),
\end{aligned}$$

where in the second line we used the fact that, by a symmetry argument, the sum of $\bar{y}_i$ is the sum of $\binom{n}{p}\times(n-p)$ terms, divided by $(n-p)$. As each y_i is treated equally this sum must be $\binom{n}{p}\times N^{-1}$ times the sum of all y's, or $\binom{n}{p}$ times $\bar{y}$. We can also simplify the first term: we have it is the sum of $\binom{n}{p}\times(n-p)^2$ terms divided by $(n-p)^2$. Each of these terms is of the form $y_{i_1}\cdot y_{i_2}$, where $i_1 = i_2$ in $\binom{n}{p}\times(n-p)$ cases and $i_1 \neq i_2$ in the remaining $\binom{n}{p}\times(n-p)^2(n-p-1)$. By the same argument

we can identify the entire summation as a multiple of the summations of the product $y_{i_1} \cdot y_{i_2}$ in both the different and equal cases. Thus, we obtain

$$\begin{aligned}
\sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \|\bar{y}_i - z\|^2 &= p\left(\binom{n}{p}\left(\frac{1}{(n-p)n}\sum_{i=1}\|y_i\|^2 + \frac{n-p-1}{(n-p)n(n-1)}\sum_{i\neq j} y_i \cdot y_j\right) - 2\binom{n}{p}\bar{y}\cdot z + \binom{n}{p}\|z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{1}{(n-p)n}\sum_{i=1}\|y_i\|^2 + \frac{n-p-1}{(n-p)n(n-1)}\sum_{i\neq j} y_i \cdot y_j - 2\bar{y}\cdot z + \|z\|^2\right) \\
&= p\binom{n}{p}\left(\left(\frac{1}{n(n-p)}\left(\sum_{i=1}\|y_i\|^2 + \frac{n-p-1}{(n-1)}\sum_{i\neq j} y_i \cdot y_j\right) - \|\bar{y}\|^2\right) + \|\bar{y} - z\|^2\right) \\
&= p\binom{n}{p}\left(\sum_{i=1}\|y_i\|^2\left(\frac{1}{n(n-p)} - \frac{1}{n^2}\right) + \sum_{i\neq j} y_i \cdot y_j\left(\frac{n-p-1}{n(n-p)(n-1)} - \frac{1}{n^2}\right) + \|\bar{y} - z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{p}{(n-p)n^2}\left(\sum_{i=1}\|y_i\|^2 - \frac{1}{n-1}\sum_{i\neq j} y_i \cdot y_j\right) + \|\bar{y} - z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{p}{(n-p)n} S_y + \|\bar{y} - z\|^2\right).
\end{aligned}$$

Linear coefficient we have

$$\begin{aligned}
\sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} (\bar{y}_i - z)\cdot\left(\tilde{y}_j^{(i)} - z\right) &= \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \bar{y}_i \cdot \tilde{y}_j^{(i)} - \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \bar{y}_i \cdot z - \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \tilde{y}_j^{(i)} \cdot z + p\binom{n}{p}\|z\|^2 \\
&= \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \bar{y}_i \cdot \tilde{y}_j^{(i)} - p\binom{n}{p}\bar{y}\cdot z - \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \tilde{y}_j^{(i)} \cdot z + p\binom{n}{p}\|z\|^2. \\
&= \sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} \bar{y}_i \cdot \tilde{y}_j^{(i)} - 2p\binom{n}{p}\bar{y}\cdot z - p\binom{n}{p}\|z\|^2.
\end{aligned}$$

In the second to last line we used the same argument as for the quadratic coefficient, and in the last line we have used the same argument but now with the $\tilde{y}$'s. For the first double sum we used the fact that there are $\binom{n}{p} \times p \times (n-p)$ terms, divided by $(n-p)$, and that all of the terms in the sum are of the form $y_{i_1} \times y_{i_2}$ with $i_1 \neq i_2$. Therefore, by symmetry, the entire sum must add up to a multiple of the sum of all such cross products. Then,

$$\begin{aligned}
\sum_{i=1}^{\binom{n}{p}} \sum_{j=1}^{p} (\bar{y}_i - z)\cdot\left(\tilde{y}_j^{(i)} - z\right) &= p\binom{n}{p}\left(\frac{1}{n(n-1)}\sum_{i\neq j} y_i \cdot y_j - 2\bar{y}\cdot z - \|z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{1}{n(n-1)}\sum_{i\neq j} y_i \cdot y_j - \|\bar{y}\|^2 + \|\bar{y}\|^2 - 2\bar{y}\cdot z - \|z\|^2\right) \\
&= p\binom{n}{p}\left(-\frac{S_y}{n} + \|\bar{y} - z\|^2\right),
\end{aligned}$$

where in the last line we used that

$$S_y = \frac{1}{n}\sum_{i=1}\|y_i\|^2 - \frac{1}{n(n-1)}\sum_{i\neq j} y_i \cdot y_j.$$

Constant term Now we expand

$$\begin{aligned}
\sum_{i=1}^{\binom{n}{p}}\sum_{j=1}^{p}\|\tilde{y}_j^{(i)} - z\|^2 &= \sum_{i=1}^{\binom{n}{p}}\sum_{j=1}^{p}\|\tilde{y}_j^{(i)}\|^2 - 2p\binom{n}{p}\bar{y}\cdot z + p\binom{n}{p}\|z\|^2 \\
&= p\binom{n}{p}\left(\frac{1}{n}\sum_{i=1}^{n}\|y_i\|^2 - 2\bar{y}\cdot z + \|z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{1}{n}\sum_{i=1}^{n}\|y_i\|^2 - \|\bar{y}\|^2 + \|\bar{y} - z\|^2\right) \\
&= p\binom{n}{p}\left(\frac{n-1}{n}S_y + \|\bar{y} - z\|^2\right).
\end{aligned}$$

The first line follows from the same arguments as before. The second line follows from the fact that the first term is a sum of $\binom{n}{p} \times p$ terms, each of them of the form $\|y_i\|^2$, and by the same symmetry argument. The last line is simply the definition of S_y.

It only remains to show (11). This is a straightforward consequence of (22). □

A.2. Proof of proposition 2

Proof. We will study first-order conditions to equation (10). Consider the function

$$f_{m,A,B,C}(\lambda) = d\log\left(\sigma_y^2\left(1 + \frac{\lambda}{m}\right)\right) + \frac{m\left(A\lambda^2 + B\lambda + C\right)}{\sigma_y^2\left(m + \lambda\right)}. \tag{23}$$

Its derivative with respect to λ, f' is given by

$$f'_{m,A,B,C}(\lambda) = \frac{mA\lambda^2 + (2m^2A + d\sigma_y^2)\lambda + m\left(mB - C + d\sigma_y^2\right)}{\sigma_y^2(m+\lambda)^2} \tag{24}$$

Note that the numerator is another quadratic function of λ with the assignments

$$\begin{aligned}
A' &\stackrel{\text{def}}{=} mA, \\
B' &\stackrel{\text{def}}{=} 2m^2A + d\sigma_y^2, \\
C' &\stackrel{\text{def}}{=} m(mB - C + d\sigma_y^2).
\end{aligned} \tag{25}$$

In our case, the values of m, A, B, C are given by

$$\begin{aligned}
m &= n - p, \\
A &= \|\bar{y} - z\|^2 + \frac{p}{n(n-p)}S_y, \\
B &= -2\left(\|\bar{y} - z\|^2 - \frac{S_y}{n}\right), \\
C &= \|\bar{y} - z\|^2 + \frac{(n-1)S_y}{n},
\end{aligned}$$

and therefore,

$$\begin{aligned}
A' &= (n-p)\|\bar{y}-z\|^2 + \frac{p}{n}S_y = (n-p)\tilde{\Delta} \\
B' &= 2(n-p)^2\|\bar{y}-z\|^2 + 2\frac{p(n-p)S_y}{n} + d\sigma_y^2 = 2\,(n-p)^2\,\tilde{\Delta} + d\sigma_y^2 \\
C' &= (n-p)\left(-\left(2(n-p)+1\right)\Delta + \frac{(2(n-p)-(n-1))S_y}{n} + d\sigma_y^2\right).
\end{aligned}$$

Likewise, the second derivative of f is given by

$$\begin{aligned}
f''_{m,A,B,C}(\lambda) &= \frac{\lambda(2A'm - B') + B'm - 2C'}{(m+\lambda)^3\sigma_y^2} \\
&= \frac{-\lambda\sigma_y^2 + B'm - 2C'}{(m+\lambda)^3\sigma_y^2}.
\end{aligned} \tag{26}$$

A sufficient condition for this second derivative to be positive in $[0,1]$ is that

$$B'm - 2C' \geq \sigma_y^2. \tag{27}$$

Suppose that (27) holds. Then, by (24), and since the condition $\sqrt{B'^2 - 4A'C'} - B' \leq 2A'$ is equivalent to $A' + B' + C' \geq 0$, we can characterize $\hat{\lambda}$ minimizing f in $[0,1]$ as

$$\hat{\lambda} = \begin{cases} \frac{1}{2A'}\left(\sqrt{B'^2 - 4A'C'} - B'\right) & \text{if } C' < 0, A' + B' + C' \geq 0', \\ 1 & \text{if } C' < 0, A' + B' + C' \leq 0', \\ 0 & \text{if } C' \geq 0. \end{cases} \tag{28}$$

Furthermore, the case $A' + B' + C' \leq 0$ can never occur. Indeed,

$$\begin{aligned}
A' + B' + C' &= m\Delta + \frac{mp}{nm}S_y + 2m^2\Delta + \frac{2m^2p}{nm}S_y + d\sigma_y^2 + m\left(-\left(2m+1\right)\Delta + \frac{(2m-(n-1))S_y}{n} + d\sigma_y^2\right) \\
&= (m+1)\sigma_y^2 + \frac{S_y}{n}\left(p + 2mp + 2m^2 - (n-1)m\right) \\
&= (n-p+1)\left(\sigma_y^2 + S_y\right)
\end{aligned}$$

We note that a sufficient condition for (27) to hold is that $C' < 0$, since $mB' \geq m\sigma_y^2$. In that case, the optimal $\hat{\lambda}_{loo}$ is given by

$$\hat{\lambda}_p = \frac{1}{2A'}\left(\sqrt{B'^2 - 4A'C'} - B'\right) \tag{29}$$

In the case $C' < 0$ then f_λ doesn't have a stationary point in $[0,1]$ so the optimal value must be either 0 or 1. We show in that case, it must be. Indeed, the we have

$$
\begin{aligned}
f_{m,A,B,C}(0) - f_{m,A,B,C}(1) &= -d\log\left(1+\frac{1}{m}\right) + \frac{m}{\sigma_y^2}\left(\frac{C}{m} - \frac{A+B+C}{m+1}\right) \\
&= d\log\left(1+\frac{1}{m}\right) + \frac{1}{\sigma_y^2(m+1)}\left(C - m(A+B)\right) \\
&= -d\log\left(1+\frac{1}{m}\right) + \frac{1}{\sigma_y^2(m+1)}\left(\Delta(1+m) + \frac{S_y}{n}\left(n-1-p-2m\right)\right) \\
&= -d\log\left(1+\frac{1}{m}\right) + \frac{1}{\sigma_y^2}\left(\Delta - \frac{S_y}{n}\right) \\
&\leq -d\log\left(1+\frac{1}{m}\right) + \frac{d}{2m+1} \\
&\leq 0.
\end{aligned}
$$

In the second to last tine we used that $C' > 0$ implies that

$$
\Delta \leq \frac{S_y}{n}\left(1 - \frac{n+1}{2(m+1)}\right) + \frac{d\sigma_y^2}{2m+1} \leq \frac{S_y}{n} + \frac{d\sigma_y^2}{2m+1}.
$$

In the last line we used that the function $f(x) = 1/(2x+1) - \log(1+1/x)$ is smaller than zero if $x \geq 1$. To see this, note that $f(1) = \log(2) - 1/3 < 0$ and $\lim_{x\to\infty} f(x) = 0$. Also, f is increasing since

$$
f'(x) = \frac{1}{x} - \frac{1}{x+1} - \frac{2}{(2x+1)^2} \geq \frac{1}{x} - \frac{1}{2x} - \frac{1}{8x^2} \geq \frac{4x-1}{8x^2} > 0.
$$

In summary, if $C' > 0$ then $\hat{\lambda}_p$ is given by 29 and otherwise $\hat{\lambda}_p = 0$. Therefore, $\hat{\lambda}_p$ is the positive part of the estimator in 29.

In what follows we will work out a more explicit expression for $\hat{\lambda}_p$, assuming it is positive. Call

$$
c_1 = (n-p), c_2 = \frac{p(n-p)}{n}, c_3 = \frac{(2(n-p)-(n-1))}{n}
$$

We have

$$
\begin{aligned}
B'^2 - 4A'C' &= \left(2c_1^2\Delta + 2c_2S_y + d\sigma_y^2\right)^2 - 4\left(-\left(2c_1+1\right)\Delta + c_3S_y + d\sigma_y^2\right)\left(c_1^2\Delta + c_2S_y\right) \\
&= \Delta^2\left(4c_1^4 + 4c_1^2(2c_1+1)\right) + 4S_y^2\left(c_2^2 - c_2c_3\right) + 4d\sigma_y^2\Delta(c_1^2 - c_1^2) \\
&+ 4d\sigma_y^2S_y\left(c_2 - c_2\right) + 4S_y\Delta\left(2c_1^2c_2 - c_1^2c_3 + c_2(2c_1+1)\right) + d^2\sigma_y^4 \\
&= 4c_1^2\Delta^2\left(c_1+1\right)^2 + 4S_y^2\left(c_2^2 - c_2c_3\right) + 4S_y\Delta\left(2c_1^2c_2 - c_1^2c_3 + c_2(2c_1+1)\right) + d^2\sigma_y^4
\end{aligned}
$$

Equation (12) follows if we are able to establish the relationships

$$
c_2^2 - c_2c_3 = \kappa_{n,p}\left(\kappa_{n,p} - 1\right), \tag{30}
$$

$$
c_1^2c_2 - c_1^2c_3 + c_2(2c_1+1) = \gamma_{n,p}(2\kappa_{n,p} - 1). \tag{31}
$$

To see (30), note that

$$\begin{aligned}
c_2^2 - c_2 c_3 &= c_2(c_2 - c_3) \\
&= \frac{p(n-p)}{n^2}\left(p(n-p) - (n-p) - (p-1)\right) \\
&= \frac{p(n-p)}{n^2}\left((n-p)(p-1) - (1-p)\right) \\
&= \frac{p(n-p)}{n^2}(n-p+1)(p-1) \\
&= \kappa_{n,p}\left(\kappa_{n,p} - 1\right).
\end{aligned}$$

Likewise, to see (31) call $A = (n-p+1) = c_1 + 1$ and note that

$$\begin{aligned}
c_1^2 c_2 - c_1^2 c_3 + c_2(2c_1+1) &= \frac{c_1}{n}\left(2(p-1)c_1^2 + c_1(n+2p-1) + p\right) \\
&= \frac{c_1}{n}\left(2(p-1)(A-1)^2 + (A-1)(2A - (n-4p+3)) + p\right) \\
&= \frac{c_1}{n}\left(2pA^2 + A\left(-4(p-1) - (n-4p+3) - 2\right)) + 2(p-1) + (n-4p+3) + p\right) \\
&= \frac{c_1}{n}\left(2pA^2 - A\left(n+1\right) + n - p + 1\right) \\
&= \frac{c_1}{n}\left(2pA^2 - A\left(n+1\right) + A\right) \\
&= c_1 A\left(\frac{2pA}{n} - 1\right) \\
&= \gamma_{n,p}(2\kappa_{n,p} - 1)
\end{aligned}$$

Note that, by a simple rearrangement of terms, we may also write $B'^2 - 4A'C'$ as

$$B'^2 - 4A'C' = \left(2\left(n-p\right)\left(n-p+1\right)\tilde{\Delta} - d\sigma_y^2\right)^2 - 4\left(n-p\right)\left(n-p+1\right)\tilde{\Delta}\left(d\sigma_y^2 - S_y\right)$$

□

A.3. Proof of proposition 3

Proof. We have

$$\begin{aligned}
R(\hat{\theta}, \theta) &= E((\hat{\theta} - \theta)'(\hat{\theta} - \theta)) \\
&= E\left((Y-\theta)'(Y-\theta)\right) - 2E\left((Y-\theta)'(Y-z)\frac{c(||Y-z||^2)}{||Y-z||^2}\right) + E\left(\frac{c^2(||Y-z||^2)}{||Y-z||^2}\right) \\
&= tr(\Sigma) - 2E\left((Y-\theta)'(Y-z)\frac{c(||Y-z||^2)}{||Y-z||^2}\right) + E\left(\frac{c^2(||Y-z||^2)}{||Y-z||^2}\right)
\end{aligned}$$

We now show that, by Stein's identity

$$E\left((Y-\theta)'(Y-z)\frac{c(||Y-z||^2)}{||Y-z||^2}\right) = E\left(\frac{c(||Y-z||^2)}{||Y-z||^2}\left(\sigma^2 d - 2\sigma^2\right)\right) + 2\sigma^2 E\left(c'(||Y-z||^2)\right).$$

Indeed, consider the vector function

$$\gamma(y) = \frac{(y-z)c(||y-z||^2)}{||y-z||^2}.$$

The gradient of its $i-th$ component is given by (e_i is the $i-th$ canonical vector)

$$\nabla\gamma_i(y) = 2(y_i - z_i)(y - z)\frac{c'(||y-z||^2)}{||y-z||^2} - 2(y_i - z_i)(y - z)\frac{c(||y-z||^2)}{||y-z||^4} + e_i\frac{c(||y-z||^2)}{||y-z||^2}.$$

Then, by Stein's lemma (Fourdrinier et al., 2018)

$$\begin{aligned} E\left((Y-\theta)'(Y-z)\frac{c(||Y-z||^2)}{||Y-z||^2}\right) &= \sigma^2 E\left(\nabla\cdot\gamma(Y)\right) \\ &= \sigma^2\sum_i 2E\left((y_i - z_i)^2\left(\frac{c'(||Y-z||^2)}{||Y-z||^2} - \frac{c(||Y-z||^2)}{||Y-z||^4}\right)\right) \\ &\quad + tr(\Sigma)E\left(\frac{c(||Y-z||^2)}{||Y-z||^2}\right) \\ &= \sigma^2\left(2c'(||Y-z||^2) + (d-2)\frac{c(||Y-z||^2)}{||Y-z||^2}\right). \end{aligned}$$

□

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